

Reader Supplement to
*Generic Identifiability and Directed Containment for Strongly
Tree-Child Level-2 Networks under the Kimura Two-Parameter
Model:*
The Principal Positive Domain and Strict Continuous Time

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Abstract

This supplement is a reader-oriented map from the mathematical proof to the exact finite certificate package. It separates hand proofs from the bounded computer-assisted residue, defines the primitive completion grammar and raw relation object, records all censuses and restoration/probe invariants, gives representative certificate schemas and the complete weak-class witness, and lists the minimal artifacts and commands needed to replay each theorem layer. The frozen principal-domain computational-evidence lock remains unchanged; the article and this supplement are separately versioned submission sources.

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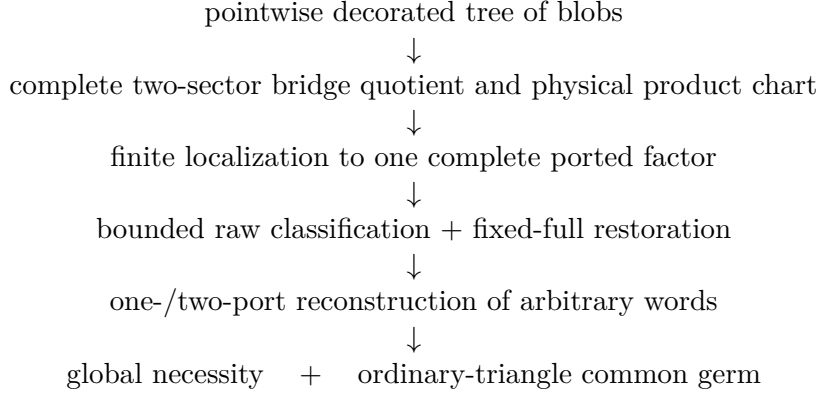
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1 How to read the proof

The main article proves a global theorem from three qualitatively different types of input.

- P1. Analytic and algebraic proofs.** These include the physical K2P domain, subdivision and rerooting, pointwise quartet and whole-map $\mathcal{T}_i$ signs, the two-sector bridge fibre, physical local products, marginal submersions, localization, genericity, continuous-time gluing, and the weak-class construction.
- P2. Topology-only finite reduction.** The cycle/theta primitive theorem and minimum-repair grammar are graph-theoretic. They reduce an arbitrary strong level-2 factor to a finite rigid support plus ordered labelled words on finitely many directed segments.
- P3. Exact bounded computation.** Raw four- and five-port directions, rank exclusions, direct polynomial separators, restoration children, and coherent probes are exhausted by exact graph and rational/symbolic arithmetic. This layer has independent regeneration and mutations, but is not presented as a complete short hand classification.

The logical dependency is



The quartet and $\mathcal{T}_i$ inequalities are pointwise. The polynomial and rank certificates are generic non-containment statements. Restoration applies only inside one assumed fixed full relation. Keeping these quantifiers separate is essential.

2 Convention crosswalk

All graphs use the following fixed convention.

- Rooted presentations are binary directed acyclic graphs; the root is the lowest stable ancestor.
- Only arcs entering reticulations retain arrowheads.
- The root is deleted once and its two incident arcs merge into one mixed edge. No later parallel-edge or degree-two cleanup is part of formation of the full standard semi-directed graph.
- Restriction/display operations may subsequently prune dead ends and suppress unlabelled bivalent vertices; they do not change the admissible-rooting universe of the original graph.
- “Strongly tree-child” quantifies over every admissible rooting; “weakly tree-child” requires at least one tree-child rooting.
- Isomorphisms preserve labels, ordinary edges, arrowheads, and vertex roles. Ordinary triangle equivalence forgets only the reticulation choice on one three-cycle and keeps its three external arms fixed.

The K2P character order is $(0, C, G, T)$, with equal sector $\{C, T\}$ and singleton sector $\{G\}$. Every edge has Fourier vector $(1, s, g, s)$ in

$$\mathcal{D}_+ = \{0 < s < 1, 0 < g < 1, g > 2s - 1\}.$$

The strict continuous-time subdomain is

$$\mathcal{D}_{\text{CT}} = \{0 < s < 1, s^2 < g < 1\}.$$

No mixed-sign stochastic component is part of the theorem.

3 Primitive graph grammar

3.1 Core segments and minimum repairs

After removing external port arms and suppressing ordinary non-event degree-two vertices, a nontrivial strong level-2 blob is a cycle or one of four directed theta event placements. Segment indices in the last column below refer to the left-to-right segment order.

core	directed segments	minimum repair sets
cycle	$S \rightarrow X, S \rightarrow X$	$\{0\}, \{1\}$
θ_0	$S \rightarrow U, S \rightarrow V, U \rightarrow X, V \rightarrow X, U \rightarrow V$	$\{2, 3\}, \{3, 4\}$
θ_1	$S \rightarrow U, S \rightarrow X, V \rightarrow X, U \rightarrow V, U \rightarrow V$	$\{2, 3\}, \{2, 4\}$
θ_2	$S \rightarrow U, S \rightarrow V, U \rightarrow X_0, V \rightarrow X_0, U \rightarrow X_1, V \rightarrow X_1$	$\{2, 3\}, \{2, 5\}, \{3, 4\}, \{4, 5\}$
θ_3	$S \rightarrow U, S \rightarrow X_0, V \rightarrow X_0, U \rightarrow X_1, V \rightarrow X_1, U \rightarrow V$	$\{2\}, \{4\}$

An occupied segment contains at least one port-bearing subdivision. The minimum repairs are the minimal occupancy sets satisfying the no-omnian and simple-graph requirements. A complete factor additionally retains one child role below every path-sink reticulation. Here “occupied” includes a named zero-character completion role inserted by the compiler; it need not be a selected physical port. Arbitrary physical factors arise by replacing each selected segment word by an ordered word of distinct labelled ports.

3.2 Completion grammar and the numbers 831, 1983, and 4155

For each core the compiler enumerates:

- (i) every allowed subset of path-sink child roles;
- (ii) every weak composition of the remaining labelled port roles among the directed segments;
- (iii) for a theta target, every minimum-repair tag; for a cycle target, the single untagged target convention (the two repaired cycle supports occur on the source side);
- (iv) the selected-incoming mode, in which a real incoming boundary remains a port, and the marginalized-incoming mode, in which that role has been omitted; and
- (v) every physical port permutation.

The directed completion grammar yields 831 four-port target descriptors in selected-incoming mode and 1,983 in marginalized-incoming mode. Five-port targets number 1,983 and 4,155 in the corresponding modes. These are repair-tagged descriptors, not necessarily distinct untagged mixed graphs. The compiler first weakly distributes the selected physical labels. For a chosen theta repair it then inserts one deterministically named zero-character leaf on each required segment whose physical word is empty, and inserts nothing extra on an already occupied required segment. The repair tag stays in the raw identity even when another repair witness yields the same final mixed graph. Unselected sink roles and a marginalized incoming role receive their own deterministic dummies. Consequently these roles consume no selected labels and the stars-and-bars count in the article is unchanged. The numbers are taken before source multiplication and port permutations. Thus

$$6(831 + 1983)4! = 405,216, \quad 4(1983 + 4155)5! = 2,946,240.$$

3.3 Raw relation schema

A raw directional relation is not an unlabelled topology hash. Its immutable identity includes:

- source primitive and exact repaired support;
- target primitive, completion descriptor, and selected graph;
- source and target incoming roles;
- physical port permutation and port match;
- omitted source and target roles;
- source-to-target direction;
- boundary incidence transport; and
- exact graph fingerprints.

Only after every raw ID is generated exactly once are presentations grouped into canonical terminal or restoration classes.

4 Certificate decision tree

The classifier uses proof types in a fixed precedence order.

type	mathematical implication	exact stored evidence
Displayed quartet	physical images are disjoint pointwise	distinct complete switching-split sets, selected witness split, and the appropriate leaf-permuted F or G polynomial.
Whole-map $\mathcal{T}_i$	physical images are disjoint pointwise	coordinate triple and orientation, exact whole-map pullbacks, and a strict Bernstein/sign certificate for the factorized pullback.
Directed rank	no full-dimensional source germ enters the target	nonzero exact source minor, target lower minor, symbolic target rank upper certificate, and strict source-rank $>$ target-rank comparison.
Polynomial separator	generic source-to-target containment is impossible	sparse integer polynomial $P(q)$, coefficientwise proof $P \circ \Phi_T = 0$, nonzero source pullback, and strict rational source witness.
Restoration parent	deferred fixed-full obligation	parent relation, each actual physical child attachment, exact parent-child transport restriction, and a terminal certificate at every leaf.
Isomorphism	physical parameter transport exists	exact labelled mixed-incidence isomorphism and parameter/port transport.
Ordinary triangle	one common full-dimensional regular germ exists	stored triangle edge set and coherent transport, combined with the exact rank-nine analytic construction in the article.

The target rank upper certificate is symbolic; a one-point Jacobian rank is never used as an upper bound. Polynomial witnesses are checked on the graph- generated K2P maps rather than accepted by a topology identifier.

5 Finite census and current compression boundary

5.1 Authoritative raw partitions

category	four-port theta	five-port θ_2
displayed quartet	360,408	2,942,592
whole-map $\mathcal{T}_i$	16,974	2,528
directed exact rank	23,822	800
direct terminal / quadratic	1,472	240
restoration member / isomorphism	2,540	80
total	405,216	2,946,240

The 1,472 four-port terminal presentations form 934 classes:

839 quadratics + 36 direct polynomials + 4 hard bindings + 20 isomorphisms + 35 triangles.

The 36 direct polynomials have three certified source-family tags: 22 θ_0 quintic port-orbit obstructions, 12 lower-theta quartics, and two θ_3 cubics. The 839 quadratic classes have eight literal certificate bodies. These repetitions make further theorem-template compression plausible, but equality of certificate bodies is not by itself a proof that all graph transports belong to one legal orbit.

5.2 Audited proof-compression table

The bounded compression pass has status **PC-PARTIAL**. It changes no theorem statement and discards no exact ledger. In particular, literal equality of polynomial bodies is not treated as a graph orbit, and the 297 restoration fingerprints are descriptive archetypes rather than a proved transport quotient.

layer	frozen units	compact units	exact residue retained
four-port quadratics	839	8	every coordinate transport
five-port θ_2 quadratics	96	4	every coordinate transport
three-port cycle quadratics	54	6	every coordinate transport
restoration quadratics	6	5	every coordinate transport
direct-36 high degree	36	27	all directional bindings; 3 certified proposition families, 5 stored base formulas
rank upper certificates	4,379	3,515+75	universal mechanism plus 75 exceptional representatives covering 864 descriptors
restoration parents	997	297	997 assignments, 2,540 roots, 36,824 edges, and 16 algebra-transport classes

The direct-36 row must not be read as three literal polynomials. The quintic, quartic, and cubic proposition families have 27 direction-specific bodies across the 36 records. The quadratic rows group only literal coordinate bodies inside their original coordinate convention or transports already certified by a frozen orbit package.

layer	uniform statement	load-bearing finite premise
primitive completions	one binomial-sum formula for all selected/marginalized cases	all labelled directed completion records; totals 405,216, 2,946,240, and 13,440
restoration	four terminal mechanisms and one depth-two continuation mechanism	the exact 997-parent assignment and every physical parent-child transport
probe reconstruction	one-/two-port coherence implies arbitrary attachment-word recovery	176 anchors, 29,964 one-port rows, 544,571 two-port rows, 67,741 transports, and 4,379 restrictions

The machine-readable theorem-to-artifact crosswalk is `proof_compression_submission/crosswalk/THEOREM_ARTIFACT_CROSSWALK.json`. It records, for each theorem layer, exact file hashes and declared schemas, producer, replay, mutation, and environment bindings. The original frozen principal-domain computational-evidence lock does not itself contain an end-to-end timing record. For C01–C10 and C12–C13, the theorem-layer crosswalk leaves end-to-end runtimes null, preserving locked component observations without summing them. Separately, its global C11 entry and the submission bundle bind a detached clean-checkout replay of the unchanged lock: all 35 layers passed in 5,172.89 seconds wall time. No byte-bound end-to-end quick-suite runtime is claimed.

5.3 Restoration forests

The corrected four-port restoration forest has 997 canonical parents and 2,540 physical member roots. Its first layer is

quartet separator	35,758	
whole-map $\mathcal{T}_i$	606	
quadratic	148	(36,568 first children).
transported quartic	24	
continuation	32	

The 32 continuations have 256 second children: 248 quartet and eight $\mathcal{T}_i$. Hence there are 36,824 edges, 36,792 exact separator leaves, maximum depth two, no cycle, no missing child, and no unresolved record. Every child transport restricts to its parent’s boundary transport.

The five-port θ_2 dummy forest has 56 roots and 864 one-parent descendants. Its 832 leaves are 760 quartet exclusions and 72 labelled isomorphisms. The cycle layer has 5,964 restoration roots and 536,364 physical completions, of which 535,920 are quartet-separated, 300 are $\mathcal{T}_i$ -separated, 132 have exact quadratics, and 12 are isomorphic.

The logical implication is always fixed-full. An assumed full relation fixes the omitted physical label, source edge, target edge, role, and transport. That actual completion is one child in the forest. Source marginal openness then turns its separator into a contradiction. No abstract parent relation is lifted, and no target deletion map is inverted.

6 Coherent probes and arbitrary subdivision words

The equality-anchor input contains 176 physical relations:

origin	ports	records
four-port direct terminals	4	26
restored four-port terminals	5	17
θ_2 direct anchors	5	24
θ_2 one-role restorations	6	40
θ_2 two-role paths	7	32
cycle direct anchors	3	24
cycle restored anchors	4	12
ordinary tree identity	3	1
total		176

There are 143 isomorphism and 33 triangle anchors.

If a rooted binary factor has k boundary leaves and r reticulations, degree counting gives $t = k + r - 2$ nonroot tree vertices and $2k + 3r - 2$ rooted arcs. Root suppression merges two arcs, leaving exactly

$$2k + 3r - 3$$

physical attachment sites. The probe input contains every site: the root-suppressed incoming segment, every other pendant arm, both incoming edges at every reticulation, and every remaining unheaded core edge. The two artificial root halves are identified only after exact semi-directed isomorphism.

Across all anchors there are 2,206 source sites and 2,206 target sites. The first-probe universe is the per-anchor Cartesian product and contains 29,964 directions. Its census is

$$27,758 \text{ quartet} + 99 \mathcal{T}_i + 1,915 \text{ isomorphism} + 192 \text{ triangle}.$$

The 2,107 equality survivors generate every two-port parent exactly once. The second-probe universe has 544,571 directions:

$$511,266 \text{ quartet} + 576 \mathcal{T}_i + 30,969 \text{ isomorphism} + 1,760 \text{ triangle}.$$

One-port restrictions identify the directed core segment containing each omitted boundary. Two-port restrictions determine the order of every pair on one segment. Since the comparisons come from a genuine word, they are transitive and recover a unique total order. The exact child transports agree on overlaps and restrict to their parents. All 33 triangle anchors transport the same inherited ordinary triangle; probes create no new triangle. This is the finite mechanism that extends the bounded theorem from rigid supports to arbitrary complete labelled subdivision words.

7 Representative exact calculations

7.1 Tree-sunlet factorization

For the three-sunlet orientation at leaf three,

$$q_{xyz} = a_x b_y c_z \{ \delta f_y d_z + (1 - \delta) f_x e_z \}, \quad x + y + z = 0.$$

With $X_h = q_{hh0}$, $Y_h = q_{h0h}$, $Z_h = q_{0hh}$, and $V = q_{CTG}$, exact substitution gives

$$V^2 X_g - X_s^2 Y_g Z_g = -a_s^2 b_s^2 a_g b_g c_g^2 f_s^2 \delta (1 - \delta) d_g e_g (1 - f_g)^2.$$

This identity is used on whole semi-directed maps. A rooted restriction type is not used as a topology oracle.

7.2 Bridge normalizer

For an unmarked degree- d component, the pair-anchor rows $(12), (13), (23), (14), \dots, (1d)$ have rank d and leading determinant -2 . If r_{ij} are positive anchor ratios, the incidence scales are

$$a_1 = \sqrt{r_{12}r_{13}/r_{23}}, \quad a_2 = \sqrt{r_{12}r_{23}/r_{13}}, \quad a_3 = \sqrt{r_{13}r_{23}/r_{12}}, \quad a_k = r_{1k}/a_1.$$

The formula is applied independently to the $\{C, T\}$ and G sectors.

7.3 Certificate schema for a direct polynomial

A sparse direct certificate stores

$$P(q) = \sum_m c_m \prod_{j \in m} q_j,$$

with integer coefficients and explicit Fourier-coordinate tuples. Its replay regenerates source and target polynomial maps from the two graph encodings, forms $P \circ \Phi_S$ and $P \circ \Phi_T$ over $\mathbb{Z}$, checks $P \circ \Phi_T = 0$ coefficientwise, checks a nonzero source polynomial hash, and evaluates it at a rational point satisfying every strict K2P and inheritance inequality. Degrees two, three, four, and five are stored and mutated separately so that certificate reassignment is rejected.

8 Printed exact-certificate appendix

This appendix exposes the formulas already derived by the first compression cycle; it is not a second compression claim. The 23 quadratic templates identify literal coefficient, coordinate-pair, and multidegree bodies only. Equality of displayed formulas across layers does not license any graph, port, direction, or coordinate transport. In particular, the repeated literal bodies in different layers remain different graph-derived certificates.

8.1 Coordinate dictionaries

Write the character group in the integer order $0, C, G, T$. A coordinate index is obtained by listing the zero-sum tuples, replacing each tuple by the lexicographically smaller of it and its global C/T swap, deleting duplicates, and sorting. Thus $q_i = q_{h_1 \dots h_k}$ for the tuple printed below. All 36 four-port coordinates occur in the printed quadratic or higher-degree formulas:

index	tuple	index	tuple	index	tuple	index	tuple
q_0	0000	q_1	00CC	q_2	00GG	q_3	0C0C
q_4	0CC0	q_5	0CGT	q_6	0CTG	q_7	0G0G
q_8	0GCT	q_9	0GG0	q_{10}	C00C	q_{11}	C0C0
q_{12}	C0GT	q_{13}	C0TG	q_{14}	CC00	q_{15}	CCCC
q_{16}	CCGG	q_{17}	CCTT	q_{18}	CG0T	q_{19}	CGCG
q_{20}	CGGC	q_{21}	CGT0	q_{22}	CT0G	q_{23}	CTCT
q_{24}	CTG0	q_{25}	CTTC	q_{26}	G00G	q_{27}	G0CT
q_{28}	G0G0	q_{29}	GC0T	q_{30}	GCCG	q_{31}	GCGC
q_{32}	GCT0	q_{33}	GG00	q_{34}	GGCC	q_{35}	GGGG

The following are exactly the 52 five-port indices occurring in the stored literal coordinate-pair blocks behind the printed theta2 and restoration quadratics. Of these, 46 have a nonzero displayed coefficient; six occur only in zero slots retained by the exact certificate-body normalization:

index	tuple	index	tuple	index	tuple	index	tuple
q_1	000CC	q_2	000GG	q_3	00C0C	q_4	00CC0
q_5	00CGT	q_6	00CTG	q_7	00G0G	q_8	00GCT
q_9	00GG0	q_{10}	0C00C	q_{18}	0CG0T	q_{26}	0G00G
q_{27}	0G0CT	q_{28}	0G0G0	q_{29}	0GC0T	q_{30}	0GCCG
q_{31}	0GCGC	q_{32}	0GCT0	q_{34}	0GGCC	q_{35}	0GGGG
q_{36}	C000C	q_{37}	C00C0	q_{40}	C0C00	q_{41}	C0CCC
q_{43}	C0CTT	q_{49}	C0TCT	q_{51}	C0TTC	q_{52}	CC000
q_{53}	CC0CC	q_{55}	CC0TT	q_{56}	CCC0C	q_{57}	CCCC0
q_{64}	CCT0T	q_{67}	CCTT0	q_{85}	CT0CT	q_{87}	CT0TC
q_{88}	CTC0T	q_{91}	CTCT0	q_{96}	CTT0C	q_{97}	CTTC0
q_{100}	G000G	q_{102}	G00G0	q_{103}	G0C0T	q_{105}	G0CGC
q_{109}	G0GGG	q_{110}	GC00T	q_{118}	GCG0C	q_{128}	GG0GG
q_{129}	GGC0C	q_{131}	GGCGT	q_{133}	GGG0G	q_{135}	GGGG0

8.2 The 23 literal quadratic bodies

The multidegree is ordered as $(s_0, g_0, s_1, g_1, \dots)$. “Canonical” counts exact certificate classes; “raw” counts their raw directional presentations.

ID	layer	polynomial	multidegree	canonical	raw
R4Q-01	four-port	$q_4q_{20} + q_5q_{21} - q_8q_{24}$ $- q_9q_{15} - q_9q_{17} + q_9q_{23}$	$(1, 0, 1, 1, 1, 1, 0)$	12	12
R4Q-02	four-port	$q_{10}q_{30} + q_{13}q_{29} - q_{15}q_{26}$ $- q_{17}q_{26} - q_{22}q_{27} + q_{23}q_{26}$	$(1, 1, 1, 0, 1, 0, 1, 1)$	52	84
R4Q-03	four-port	$q_{10}q_{20} - q_{12}q_{18}$	$(2, 0, 0, 1, 0, 1, 2, 0)$	252	424
R4Q-04	four-port	$q_3q_{19} - q_6q_{18} - q_7q_{23}$ $+ q_8q_{22}$	$(1, 0, 1, 1, 1, 0, 1, 1)$	24	40
R4Q-05	four-port	$q_3q_{19} + q_6q_{18} - q_7q_{23}$ $- q_8q_{22}$	$(1, 0, 1, 1, 1, 0, 1, 1)$	68	108
R4Q-06	four-port	$q_1q_{34} - q_8q_{27}$	$(0, 1, 0, 1, 2, 0, 2, 0)$	292	456
R4Q-07	four-port	$q_{11}q_{31} + q_{12}q_{32} - q_{23}q_{28}$ $- q_{24}q_{27}$	$(1, 1, 1, 0, 1, 1, 1, 0)$	8	8
R4Q-08	four-port	$q_1q_{16} - q_2q_{17} + q_5q_{13}$ $- q_6q_{12}$	$(1, 0, 1, 0, 1, 1, 1, 1)$	131	220
T2Q-01	five-port theta2	$q_2q_{133} - q_7q_{128} - q_{26}q_{109}$ $+ q_{35}q_{100}$	$(0, 1, 0, 1, 0, 1, 0, 1, 0, 2)$	24	60
T2Q-02	five-port theta2	$q_2q_{135} - q_9q_{128} - q_{28}q_{109}$ $+ q_{35}q_{102}$	$(0, 1, 0, 1, 0, 1, 0, 2, 0, 1)$	24	60
T2Q-03	five-port theta2	$q_{36}q_{91} + q_{37}q_{64} - q_{40}q_{87}$ $- q_{49}q_{52}$	$(2, 0, 1, 0, 1, 0, 1, 0, 1, 0)$	16	40
T2Q-04	five-port theta2	$q_{36}q_{57} - q_{36}q_{67} - q_{37}q_{56}$ $+ q_{37}q_{88} + q_{40}q_{55} - q_{40}q_{87}$ $- q_{43}q_{52} + q_{49}q_{52}$	$(2, 0, 1, 0, 1, 0, 1, 0, 1, 0)$	32	80
C3Q-01	three-port cycle in four-port coordinates	$q_4q_{20} + q_5q_{21} - q_8q_{24}$ $- q_9q_{15} - q_9q_{17} + q_9q_{23}$	$(1, 0, 1, 1, 1, 1, 1, 0)$	2	4
C3Q-02	three-port cycle in four-port coordinates	$q_4q_{30} - q_6q_{32}$	$(0, 1, 2, 0, 2, 0, 0, 1)$	16	40
C3Q-03	three-port cycle in four-port coordinates	$q_1q_{34} - q_8q_{27}$	$(0, 1, 0, 1, 2, 0, 2, 0)$	16	40
C3Q-04	three-port cycle in four-port coordinates	$q_1q_{16} - q_2q_{17} - q_5q_{13}$ $+ q_6q_{12}$	$(1, 0, 1, 0, 1, 1, 1, 1)$	2	4
C3Q-05	three-port cycle in four-port coordinates	$q_1q_{16} - q_2q_{17} + q_5q_{13}$ $- q_6q_{12}$	$(1, 0, 1, 0, 1, 1, 1, 1)$	2	4
C3Q-06	three-port cycle in four-port coordinates	$q_{11}q_{19} - q_{13}q_{21}$	$(2, 0, 0, 1, 2, 0, 0, 1)$	16	40
RSQ-01	five-port restoration	$q_3q_{131} - q_5q_{129} - q_{29}q_{105}$ $+ q_{31}q_{103}$	$(0, 1, 0, 1, 2, 0, 0, 1, 2, 0)$	1	1

ID	layer	polynomial	multidegree	canonical	raw
RSQ-02	five-port restoration	$q_1 q_{34} - q_8 q_{27}$	$(0, 0, 0, 1, 0, 1, 2, 0, 2, 0)$	2	6
RSQ-03	five-port restoration	$q_4 q_{30} - q_6 q_{32}$	$(0, 0, 0, 1, 2, 0, 2, 0, 0, 1)$	1	10
RSQ-04	five-port restoration	$q_{10} q_{118} - q_{18} q_{110}$	$(0, 1, 2, 0, 0, 1, 0, 0, 2, 0)$	1	130
RSQ-05	five-port restoration	$q_3 q_{131} - q_5 q_{129} + q_{29} q_{105} - q_{31} q_{103}$	$(0, 1, 0, 1, 2, 0, 0, 1, 2, 0)$	1	1

8.3 Five high-degree bases and their licensed transports

Only the 22 port permutations, 12 quartic coordinate transports, and two cubic record bindings printed here are licensed. No other port, source-target, inheritance, pole/sink, or C/T transport is inferred.

HDQ-01: theta0 quintic. Degree 5, multidegree $(2, 1, 1, 1, 2, 1, 1, 1)$.

$$\begin{aligned}
F_{HDQ-01} = & q_0 q_0 q_{11} q_{17} q_{35} - q_0 q_0 q_{11} q_{25} q_{35} - 2q_0 q_0 q_{17} q_{19} q_{28} + 2q_0 q_0 q_{19} q_{25} q_{28} \\
& - 2q_0 q_1 q_{11} q_{14} q_{35} + 2q_0 q_1 q_{13} q_{24} q_{33} + 4q_0 q_1 q_{14} q_{19} q_{28} - 2q_0 q_1 q_{21} q_{24} q_{26} \\
& + 2q_0 q_2 q_{10} q_{21} q_{32} + q_0 q_2 q_{11} q_{17} q_{33} - 2q_0 q_2 q_{11} q_{21} q_{29} - 2q_0 q_2 q_{11} q_{23} q_{33} \\
& + q_0 q_2 q_{11} q_{25} q_{33} - 2q_0 q_2 q_{14} q_{21} q_{27} + 2q_0 q_4 q_{10} q_{11} q_{35} - 4q_0 q_4 q_{10} q_{19} q_{28} \\
& - 2q_0 q_4 q_{12} q_{13} q_{33} + 2q_0 q_4 q_{12} q_{21} q_{26} - 2q_0 q_5 q_{11} q_{13} q_{33} + 2q_0 q_5 q_{11} q_{21} q_{26} \\
& + q_0 q_7 q_{11} q_{17} q_{28} - q_0 q_7 q_{11} q_{25} q_{28} - 2q_0 q_9 q_{10} q_{13} q_{32} + 2q_0 q_9 q_{11} q_{13} q_{29} \\
& - q_0 q_9 q_{11} q_{17} q_{26} + 2q_0 q_9 q_{11} q_{23} q_{26} - q_0 q_9 q_{11} q_{25} q_{26} + 2q_0 q_9 q_{13} q_{14} q_{27} \\
& - 2q_1 q_7 q_{11} q_{14} q_{28} + 2q_2 q_3 q_{11} q_{11} q_{33} - 2q_3 q_9 q_{11} q_{11} q_{26} + 2q_4 q_7 q_{10} q_{11} q_{28}
\end{aligned}$$

HDQ-02: lower-theta quartic. Degree 4, multidegree $(3, 0, 2, 1, 1, 1, 2, 1)$.

$$\begin{aligned}
F_{HDQ-02} = & q_0 q_{15} q_{16} q_{18} + q_0 q_{16} q_{18} q_{23} - q_0 q_{20} q_{22} q_{23} + q_1 q_{14} q_{16} q_{18} \\
& - 2q_3 q_{11} q_{16} q_{18} + q_3 q_{11} q_{20} q_{22} - q_4 q_{12} q_{18} q_{22} - q_8 q_{10} q_{14} q_{16} \\
& + q_8 q_{12} q_{14} q_{22}
\end{aligned}$$

HDQ-03: lower-theta quartic. Degree 4, multidegree $(2, 1, 3, 0, 1, 1, 2, 1)$.

$$\begin{aligned}
F_{HDQ-03} = & q_0 q_{16} q_{17} q_{29} - q_0 q_{16} q_{23} q_{29} - q_0 q_{17} q_{22} q_{31} + q_0 q_{22} q_{23} q_{31} \\
& - q_1 q_{14} q_{16} q_{29} + q_1 q_{14} q_{22} q_{31} + 2q_3 q_{11} q_{16} q_{29} - q_3 q_{11} q_{22} q_{31} \\
& - q_3 q_{14} q_{16} q_{27} - q_5 q_{11} q_{22} q_{29} + q_5 q_{14} q_{22} q_{27}
\end{aligned}$$

HDQ-04: lower-theta quartic. Degree 4, multidegree $(2, 1, 3, 0, 2, 1, 1, 1)$.

$$\begin{aligned}
F_{HDQ-04} = & q_0 q_{15} q_{16} q_{32} + q_0 q_{16} q_{23} q_{32} - q_0 q_{23} q_{24} q_{30} + q_1 q_{14} q_{16} q_{32} \\
& - 2q_3 q_{11} q_{16} q_{32} + q_3 q_{11} q_{24} q_{30} - q_4 q_{14} q_{16} q_{27} - q_6 q_{10} q_{24} q_{32} \\
& + q_6 q_{14} q_{24} q_{27}
\end{aligned}$$

HDQ-05: theta3 cubic. Degree 3, multidegree $(1, 1, 1, 1, 1, 1, 1, 1)$.

$$\begin{aligned}
F_{HDQ-05} = & q_0 q_{15} q_{35} + q_0 q_{20} q_{30} + q_1 q_{14} q_{35} + q_2 q_{15} q_{33} \\
& + q_2 q_{18} q_{32} - q_2 q_{23} q_{33} - q_3 q_{11} q_{35} - q_4 q_{20} q_{26} \\
& + q_6 q_{12} q_{33} - q_6 q_{18} q_{28} - q_7 q_{12} q_{32} - q_7 q_{15} q_{28} \\
& + q_7 q_{23} q_{28} - q_9 q_{10} q_{30}
\end{aligned}$$

Quintic port transports.

class	port permutation	source terms
25	(0, 1, 3, 2)	1599
26	(0, 2, 1, 3)	953
27	(0, 2, 3, 1)	1524
28	(0, 3, 1, 2)	985
29	(0, 3, 2, 1)	96
30	(1, 0, 2, 3)	730
31	(1, 0, 3, 2)	1988
32	(1, 2, 0, 3)	953
33	(1, 2, 3, 0)	2004
34	(1, 3, 0, 2)	985
35	(1, 3, 2, 0)	868
36	(2, 0, 1, 3)	730
37	(2, 0, 3, 1)	1722
39	(2, 1, 3, 0)	1807
40	(2, 3, 0, 1)	96
41	(2, 3, 1, 0)	868
42	(3, 0, 1, 2)	1988
43	(3, 0, 2, 1)	1722
44	(3, 1, 0, 2)	1599
45	(3, 1, 2, 0)	1807
46	(3, 2, 0, 1)	1524
47	(3, 2, 1, 0)	2004

Quartic coordinate transports.

source	class	base	coordinate action	port match
2	112	HDQ-02	identity	(0, 1, 2, 3)
2	113	HDQ-03	identity	(0, 1, 3, 2)
2	114	HDQ-03	identity	(1, 0, 2, 3)
2	115	HDQ-02	identity	(1, 0, 3, 2)
3	112	HDQ-02	identity	(0, 1, 2, 3)
3	113	HDQ-03	identity	(0, 1, 3, 2)
3	114	HDQ-03	identity	(1, 0, 2, 3)
3	115	HDQ-02	identity	(1, 0, 3, 2)
4	8	HDQ-04	identity	(0, 1, 2, 3)
4	9	HDQ-04	swap_01	(0, 1, 3, 2)
4	10	HDQ-04	swap_01	(1, 0, 2, 3)
4	11	HDQ-04	identity	(1, 0, 3, 2)

Cubic record bindings.

source	class	base	port match
5	9	HDQ-05	(0, 1, 3, 2)
5	10	HDQ-05	(1, 0, 2, 3)

8.4 Three worked paths

A direct quadratic separator. Template R4Q-03 is

$$q_{10}q_{20} - q_{12}q_{18}, \quad \text{mdeg} = (2, 0, 0, 1, 0, 1, 2, 0).$$

It covers 252 canonical classes and 424 raw directions. For the representative `source_0:class_000000`, its source pullback has 44 nonzero terms and its target pullback is identically zero. The exact certificate is `d39c8b1f09360f48e9ae6e6ed55222d35cdc675fdd22c038781a7ff5e0328f5`, bound to the graph record by `10d7d0efe1d950f6f74585e2e385a5a07dfd33a2a51b56954a54b01c6b1aa44b`. This proves the stated source-to-target noncontainment; reversing the direction is not licensed.

A complete restoration root. For `s1:c658:t2792:p1032`, the following table prints every actual first- and second-child archetype. The row hash binds the role, insertion site, proof mechanism, transport fields, and evidence certificate.

layer	row	restored role	site	proof	status	row SHA-256
1	31976	D_REPAIR_O_2	0	displayed_quartet_mismatch	separated	d2a1d79aaa37de82495c0db5478793b02dc15b7500f2783506614a3afc46531d
1	31977	D_REPAIR_O_2	1	displayed_quartet_mismatch	separated	18a009d352b311939bb21650fba5d66ce0a1a8dbe29ebaa7bd6393e0c0d25748
1	31978	D_REPAIR_O_2	2	full_map_Ti_zero_strict_sign	separated	a762631aa3a30f73764549778fcd2fceb9bcaac256430648a4f4ab1bdabb4869
1	31979	D_REPAIR_O_2	3	displayed_quartet_mismatch	separated	181aad684be16b6b531b5c2a8a8cbbd5ab4492018427e921140695868c4bc29e
1	31980	D_REPAIR_O_2	4	displayed_quartet_mismatch	separated	a37508b280187028982572253c3a59826137feb537228fb443fb79b684543626
1	31981	D_REPAIR_O_2	5	displayed_quartet_mismatch	separated	d7a4960b94bf787bdc5ddeafad029506a4693b5aa5876f5ecb947500abcf1a6c
1	31982	D_REPAIR_O_2	6	displayed_quartet_mismatch	separated	d5ad83a75f35c4b1bd795d89363c7b147d10ff6629930c327b3916883cd75ba2
1	31983	INCOMING	0	displayed_quartet_mismatch	separated	8246f706cb8e56b848238ca6e8715cc7ec257691cc03f567b97cac4843ec400a
1	31984	INCOMING	1	displayed_quartet_mismatch	separated	12bba2049bad09e6403af99ae1cd8f54003aafd14654396f2328220ac954b079
1	31985	INCOMING	2	displayed_quartet_mismatch	separated	8893a4e1763de61a221d08388243e53f69e501f2e89cee01d493e6ae6eb75f95
1	31986	INCOMING	3	displayed_quartet_mismatch	separated	07b72f4608262f6fc6958a0f24c966984c5d8d70c08f490abfa216cadb2c801e
1	31987	INCOMING	4	restore_remaining_physical_role	continuation	60636f150c4eb58cd64c0126bdb1d4aafe8e48927fc798974dfba76a5c51d272
1	31988	INCOMING	5	displayed_quartet_mismatch	separated	718c7d8d3fd230a2bd1b25ecbb0faf7f59a6bb6a8b0cd261f530806b3aac25b7
1	31989	INCOMING	6	displayed_quartet_mismatch	separated	b86c1e7ab3211f299a5b9eb5910e3ef3b3ae35f69065cf189be6e31d04d288b8
2	1	D_REPAIR_O_2	0	displayed_quartet_mismatch	separated	daddc54b748d86b7f3698b771e1ea090b3fee641becc9b68fb76b0775d2b48e1

layer	row	restored role	site	proof	status	row SHA-256
2	2	D_REPAIR_O_2	1	displayed_quartet_mismatch	separated	3d678ac9082f6b02ffebbb34085488455455345b13f1e037681b75051c81a603e
2	3	D_REPAIR_O_2	2	full_map_Ti_zero_strict_sign	separated	dbe28657406837bfe91de37dc42fb124c9a71d2c475bf1655811deeeac55ed00
2	4	D_REPAIR_O_2	3	displayed_quartet_mismatch	separated	e655cf2cf0aa1572b4a83ad7b24bbe fde08e7fd968b0906c31cc3beb6977d72a
2	5	D_REPAIR_O_2	4	displayed_quartet_mismatch	separated	f61fd9ab276190e4e08a3e91288d012d46fad0e92ef586e74c9c20da566280d2
2	6	D_REPAIR_O_2	5	displayed_quartet_mismatch	separated	5145c0082ec9fc0bfc40d4315a851cf39f634e4ea3b14504f3f455a8139b0233
2	7	D_REPAIR_O_2	6	displayed_quartet_mismatch	separated	61bd0a2485ca2ba94809531aae4245dd53821a10a041f95ff1e33b1f20f118a1
2	8	D_REPAIR_O_2	7	displayed_quartet_mismatch	separated	1b825df66c3c7d4f9c9cf2156fa16cc7dd1fe9dc0849deb83d0203bfcea3a64d

There are 14 first children: 12 displayed-quartet terminals, one full-map $\mathcal{T}_i$ terminal, and one continuation (row 31987). That continuation has exactly eight children: seven displayed-quartet terminals and one full-map $\mathcal{T}_i$ terminal. Every second child has an empty remaining-role list. Hence this exact subtree has 21 leaves, depth 2, zero unresolved records, and no cycle.

A one-/two-port word transport. The anchor `tree:k3:identity` is the labelled identity tree transport `4342a65ef53537888b371724949a9ec3260731896777270eceed8fea0b475a7f`. At source and target site zero, inserting label 3 gives the exact one-port transport `c2ed6cbcd9d725664268416f315268cfdab6e2c2844790c7596aab7010e0a4e0`; both parent restrictions are `R:57a56588cb2934a8ad6d9500e0d28a8f7a3e8f797ba05e6a213af02a676eb539`. Inserting label 4 at the new site zero gives the exact two-port transport `8e3c37f3b5c32225fbf97286be01da6754a33e4fd267e4741a918740d5a4c8ba`, whose parent transport is precisely the one-port transport. Its two restriction records are `R:b286c2133804e765ea7889cbdc790218c360023201fbd93991e608af071e4bd3` and `R:0bf3e2b848168b2d5c1d689601f1c1700bd3f98502524f32763d20ddac093639`. The reversed marginal is the exact one-port transport `c144b9418ca2bc530028a613a74290e7ded74d9b38589f323cd7e01b7bdf33b6` on the same base anchor. The selected ledger rows have hashes `67691e6e518436f8911fe0dc82402de996b0deef87e3f7ad6396fb0b81bd0ef0` and `44c2b033d79916a3226985efc6211b4b54844953f90a7c00b2788a0cfe206193`. Thus both deletion orders are bound, not inferred from an unrecorded symmetry.

8.5 Provenance and replay

field	exact value
schema/version	k2p-printed-certificate-appendix-v1
producer	.env/bin/python -B proof_compression_submission/templates/build_printed_certificate_appendix.py -write
deterministic check	.env/bin/python -B proof_compression_submission/templates/build_printed_certificate_appendix.py -check
independent replay	.env/bin/python -B proof_compression_submission/templates/verify_printed_certificate_appendix.py
mutation suite	.env/bin/python -B proof_compression_submission/templates/test_printed_certificate_appendix_mutations.py
sealed payload	9d0f9abdb7b113bbe6b3e4c022d9e710ba67d8a3a8ddfe04f9de11013ebfd3aa

The authoritative input bindings are:

path	SHA-256
proof_compression_submission/templates/DIRECT_CERTIFICATE_TEMPLATE_TABLE.json	febc5e36c07e17dec1ba3dfffb70dd0e4f4030dd796225d37dd0e7cc267485fe1
work/corrected_composite_ledgers/artifacts/raw4_terminal_certificate_registry.json.gz	0a1818655429d60660c1ed87f3fbe412701f386b081562b3a4ca54079069f1d
work/theta0_quintic_orbit_certificate.json	f863afd5875a74be818141990863344fae09fd4269a803d3a1bfeb67a8a595e0
work/theta_quartic_obstruction_certificates.json	5204593fb2b47914dbdf2d7846d1e9fbd5671fa9f29e89862d29b16a45bb08db
work/theta3_cubic_obstruction_certificate.json	fb1512e260b5a88b5ac3a4b55d6c756e401baebd08866d6e39bb2153b63aa4d8
work/four_port_direct_residual_closure_certificate.json	fb6e5f1c23c8c3291ddc8c822171cae9c2df05b0d449249813b40e958e17bddc
work/restoration_sign_reclassification/corrected_restoration_forest.json	43bd2be5e7626a954fc4fa4cf45e8d0e6483c947ddc9cba80f2b1a13351bc3a8
work/probe_coherence_corrected/probe_coherence_certificate.json	93de7b0dd3aa581bdf12288eae8cb9ac42f20a9d9bb3eab35eee8ef9a759d390
work/probe_coherence_corrected/one_port_ledger.jsonl.gz	d5fa13d38731bfff2403eeb4e4d9e139566c4983b09d30553c6260eaac64c5c90
work/probe_coherence_corrected/two_port_ledger.jsonl.gz	10f0afcab77f2d61cecfcc36d723c6f32065c304ac088b0b8ecf12dfc867fbf9d
work/probe_coherence_corrected/parent_restriction_ledger.jsonl.gz	5d1e6c2fe38d31f6304a76886ec37829215b88c8b179f5b23596d49d37ceeb38
work/probe_coherence_corrected/exact_transport_ledger.jsonl.gz	6bc8e88feac2bee68491287775f078e8e5474bf930961a7390967c9fd350044d

9 Full weak-class certificate

The base graphs are encoded by

$$\begin{aligned}
W : \quad & rS, rL_0, SU, SV, UX, VZ, ZX, UV, ZL_1, XL_2, & \text{Ret} = \{V, X\}, \\
W' : \quad & rS, rL_0, SU, SX_0, VX_0, UX_1, VX_1, UV, X_0L_1, X_1L_2, & \text{Ret} = \{X_0, X_1\}.
\end{aligned}$$

Complete rooting censuses are (5, 2, 3) and (7, 2, 5) for admissible, tree-child, and non-tree-child rootings. Exact mixed-incidence comparison returns relation **none**, including after forgetting head flags on the unique triangle of each graph.

Put $\delta = 2^{-30}$. The complete parameter mapping is:

datum	W	W'
seven internal pairs	(1/7, 1/7)	(1/4, 1/4)
inheritance 1	$Z \rightarrow X : 15996/16339; U \rightarrow X : 343/16339$	$V \rightarrow X_1 : 1/2; U \rightarrow X_1 : 1/2$
inheritance 2	$S \rightarrow V : 1/8; U \rightarrow V : 7/8$	$V \rightarrow X_0 : 1/6; S \rightarrow X_0 : 5/6$
leaf 0 pair	(86779 δ /80, 86779 δ /80)	(16 δ /3, 16 δ /3)
leaf 1 pair	(320 δ /253, 320 δ /253)	(32 δ /9, 32 δ /9)
leaf 2 pair	(114373 δ /20240, 114373 δ /20240)	(96 δ /5, 96 δ /5)

In character code order 0, $C, G, T = 0, 1, 2, 3$, the common coordinate order is

$$(000, 0CC, 0GG, C0C, CC0, CGT, CTG, G0G, GCT, GG0).$$

The value is 1 at 000, δ^2 at the six two-nonzero coordinates, and $4\delta^3/5$ at the three all-nonzero coordinates.

The normalized tensors before pendant scaling are, respectively,

$$\left(1, \frac{64009}{457492}, \frac{64009}{457492}, \frac{6400}{39229939}, \frac{1}{1372}, \frac{4048}{39229939}, \frac{4048}{39229939}, \frac{6400}{39229939}, \frac{4048}{39229939}, \frac{1}{1372}\right)$$

and

$$\left(1, \frac{15}{1024}, \frac{15}{1024}, \frac{5}{512}, \frac{27}{512}, \frac{9}{4096}, \frac{9}{4096}, \frac{5}{512}, \frac{9}{4096}, \frac{27}{512}\right).$$

The graph-derived canonical orders of the visible edge classes are

$$W : (ZX, SV, rS, SU, UV, VZ, UX), \quad W' : (VX_1, VX_0, UV, rS, SX_0, SU, UX_1).$$

Accordingly, columns $0, \dots, 8$ mean

$$\begin{aligned} W &:: (s_{ZX}, g_{ZX}, s_{SV}, g_{SV}, s_{rS}, g_{rS}, s_{SU}, g_{SU}, s_{UV}), \\ W' &:: (s_{VX_1}, g_{VX_1}, s_{VX_0}, g_{VX_0}, s_{UV}, g_{UV}, s_{rS}, g_{rS}, s_{SX_0}). \end{aligned}$$

These names are a derived crosswalk, not an unstated field of the frozen certificate: the crosswalk verifier reconstructs the descendant signatures from the two primitive graphs, binds the frozen graph and descriptor hashes, and rejects a reordered column. The sealed artifact and independent checker are, respectively,

```
proof_compression_submission/analysis/WEAK_SHARPNESS_COLUMN_CROSSWALK.json
proof_compression_submission/analysis/verify_weak_sharpness_column_crosswalk.py.
```

All output-row indices in the following table are zero-based in the displayed Fourier-coordinate order; row 0 is the all-zero coordinate. The minors' rows and determinants are

map	output rows	determinant
W	(1, 2, 3, 5, 4, 7, 6, 8, 9)	10368019213741323/563981315074464023964442388464888915634290688
W'	(1, 2, 3, 5, 4, 6, 7, 8, 9)	1435825/85002596691653613846528

For cherry induction use actual physical pairs $(u_s, u_g) = (2/5, 4/9)$ and $(v_s, v_g) = (3/7, 5/11)$. The observable block $(u_s/v_s, u_s v_s, u_g/v_g, u_g v_g)$ has determinant $2464/675$. More explicitly, for an outside leaf j and new cherry leaves a, b ,

$$\frac{Q_{j=C, a=C, b=0}}{Q_{j=C, a=0, b=C}} = \frac{u_s}{v_s}, \quad Q_{a=C, b=C} = u_s v_s,$$

with all omitted characters zero; replacing C by G gives the other two observables. Square-root recovery gives the four edge factors, and division by the corresponding nonzero factor recovers every old tensor coordinate. This is the analytic local inverse; graph pruning separately preserves weak-not-strong status and inequivalence.

10 Theorem-to-artifact crosswalk

All paths below are relative to the `k2p_level2_identifiability_closure` project root. “Primary” is the producer/certificate path; “replay” is a separately implemented checker or audit. Historical files not named here are not promotion evidence. This printed table is a reader map, not the complete machine schema. The field-rich authority is `proof_compression_submission/crosswalk/THEOREM_ARTIFACT_CROSSWALK.json`, which records each claim’s declared artifact schemas, exact byte hashes, producer, replay and mutation artifacts, environment profile, and whether a runtime is known or deliberately null. Executable entry points are invoked from the project root with the pinned Python environment shown in Section 12.

claim	primary proof/certificate	independent replay or mutation
K2P domain, subdivision, rooting	work/domain_rooting_closure/PROOF.md; work/domain_rooting_closure/domain_rooting_certificate.json	work/domain_rooting_closure/verify_domain_rooting.py
Quartet signs and tree of blobs	work/quartet_separation_closure/PROOF.md; work/quartet_separation_closure/quartet_logic_certificate.json	work/quartet_separation_closure/verify_quartet_logic.py; work/adversarial_proof_review/verify_topology_direction.py
Whole-map $\mathcal{T}_i$ and triangle germ	package/original/checkpoint_2/continuation_2/K2P_TREE_SUNLET_SIGN_CERTIFICATE.md; package/original/checkpoint_2/continuation_2/K2P_TRIANGLE_GERM_EXACT.md	package/original/checkpoint_2/continuation_2/verify_triangle_and_sunlet.py; work/final_theorem_release/no_assert_triangle_sunlet.py
Bridge fibre, marginal, gluing	work/bridge_marginal_closure/PROOF.md; work/bridge_marginal_closure/certificate.json	work/bridge_marginal_closure/verify_bridge_marginal.py; work/adversarial_proof_review/PHYSICAL_LOCAL_PRODUCT_REPAIR.md; work/adversarial_proof_review/verify_adversarial.py
Global classification, genericity, reconstruction, and continuous time	work/global_theorem_closure/promotion_manuscript/K2P_SAME_PROMOTION_MANUSCRIPT.md	work/global_theorem_closure/promotion_manuscript/QUANTIFIER_AUDIT.md; work/global_proof_adversary/AUDIT.md
Four-port raw universe and terminals	work/corrected_composite_ledgers/artifacts/raw4_corrected_composite_ledger.jsonl.gz; work/corrected_composite_ledgers/artifacts/raw4_terminal_certificate_registry.json.gz	work/corrected_composite_ledgers/verify_corrected_composites_independent.py; work/corrected_composite_ledgers/run_composite_mutations.py
Five-port θ_2	work/corrected_composite_ledgers/artifacts/theta2_corrected_composite_ledger.jsonl.gz	work/corrected_composite_ledgers/verify_corrected_composites_independent.py; work/corrected_composite_ledgers/run_composite_mutations.py, family theta2
Printed direct templates and worked paths	proof_compression_submission/templates/PRINTED_CERTIFICATE_APPENDIX.json; proof_compression_submission/supplement/certificate_appendix.tex	proof_compression_submission/templates/verify_printed_certificate_appendix.py; proof_compression_submission/templates/test_printed_certificate_appendix_mutations.py

claim	primary proof/certificate	independent replay or mutation
997-parent forest	work/restoration_sign_reclassification/corrected_restoration_forest.json	work/restoration_sign_reclassification/corrected_restoration_replay_certificate.json; work/restoration_sign_reclassification/verify_corrected_restoration_forest.py; work/restoration_sign_reclassification/mutate_corrected_restoration_forest.py
Three-port cycle closure	work/cycle_three_port_closure/PROOF.md; work/cycle_three_port_closure/promotion/cycle_promotion_certificate.json	work/adversarial_proof_review/verify_corrected_cycle_promotion.py; work/adversarial_proof_review/test_cycle_promotion_mutations.py
Probe input completeness	work/adversarial_proof_review/PROBE_INPUT_CONTRACT.md; work/adversarial_proof_review/probe_input_contract.json	work/adversarial_proof_review/verify_probe_input_contract.py; work/adversarial_proof_review/test_probe_input_mutations.py
One-/two-port probe closure	work/probe_coherence_corrected/PROOF.md; work/probe_coherence_corrected/probe_coherence_certificate.json	work/probe_coherence_corrected/probe_coherence_independent_verification.json; work/probe_coherence_corrected/verify_probe_coherence_corrected.py; work/probe_coherence_corrected/run_probe_coherence_mutations.py
Primitive probe audit	work/global_proof_adversary/probe_full_audit/independent_probe_graph_audit_certificate.json	work/global_proof_adversary/probe_full_audit/independent_probe_graph_audit.py; work/global_proof_adversary/probe_full_audit/independent_probe_mutation_report.json

claim	primary proof/certificate	independent replay or mutation
Weak sharpness	work/weak_sharpness_closure/ PROOF.md; work/weak_sharpness_ closure/weak_sharpness_ certificate.json; proof_compression_submission/ analysis/WEAK_SHARPNESS_ COLUMN_CROSSWALK.json	work/weak_sharpness_audit/ PROOF_AUDIT.md; work/weak_sharpness_audit/ audit_certificate.json; work/weak_sharpness_audit/ test_mutations.py; proof_compression_submission/ analysis/verify_weak_ sharpness_column_crosswalk. py; proof_compression_submission/ analysis/test_weak_sharpness_ column_crosswalk_mutations.py
Unified theorem lock	work/final_theorem_release/ RELEASE_LOCK.json; work/final_ theorem_release/verify_final_ theorem_release.py	work/final_theorem_release/ corrected_universe_ independent_replay.json; work/final_theorem_release/ run_release_mutations.py

10.1 Printed finite-layer replay metadata

For the principal computed layers, each row below supplies the requested schema/version, authoritative file SHA-256, producer command, independent replay command, and mutation command. Commands run from the project root; none may be invoked with optimized Python.

layer	exact machine metadata
raw four-port topology ledger	schema/version: k2p-four-port-raw-ledger-summary-v1; authority: work/raw_ledger_audit/artifacts/raw_ledger_summary.json; SHA-256: 07890bfb28e8aa3f3c6883e903911e7639264d86c9c99eef3acb2223b44217c0; producer command: .venv/bin/python -B work/raw_ledger_audit/generate_raw_ledger.py; replay command: .venv/bin/python -B work/raw_ledger_audit/verify_raw_ledger.py; mutation command: .venv/bin/python -B work/raw_ledger_audit/test_mutations.py.
four-port exact-rank filter	schema/version: k2p-four-port-exact-generic-rank-upper-coverage-v1; authority: work/rank_upper_certificates/rank_upper_coverage.json; SHA-256: c52c5730494eb894360c17b6e54ae5c260fca3cddb8702d5c796750c7df874bc; producer command: PYTHONPATH=package/referee/k2p_offline_sweep_portable/atlas:work/rank_upper_certificates .venv/bin/python -B work/rank_upper_certificates/build_rank_upper_coverage.py; replay command: PYTHONPATH=package/referee/k2p_offline_sweep_portable/atlas:work/rank_upper_certificates .venv/bin/python -B work/rank_upper_certificates/verify_rank_upper_certificates.py; mutation command: PYTHONPATH=package/referee/k2p_offline_sweep_portable/atlas:work/rank_upper_certificates .venv/bin/python -B work/rank_upper_certificates/mutation_tests.py.
direct terminal templates	schema/version: k2p-printed-certificate-appendix-v1; authority: proof_compression_submission/templates/PRINTED_CERTIFICATE_APPENDIX.json; SHA-256: 7a24443f1583f1fc2ef57734c7845bc8377dc7da124265d1588f2d41e9555afa; producer command: .venv/bin/python -B proof_compression_submission/templates/build_printed_certificate_appendix.py -write; replay command: .venv/bin/python -B proof_compression_submission/templates/verify_printed_certificate_appendix.py; mutation command: .venv/bin/python -B proof_compression_submission/templates/test_printed_certificate_appendix_mutations.py.

layer	exact machine metadata
997-parent restoration forest	schema/version: k2p-corrected-restoration-forest-v3; authority: work/restoration_sign_reclassification/corrected_restoration_forest.json; SHA-256: 43bd2be5e7626a954fc4fa4cf45e8d0e6483c947ddc9cba80f2b1a13351bc3a8; producer command: .venv/bin/python -B work/restoration_sign_reclassification/build_corrected_restoration_forest.py; replay command: .venv/bin/python -B work/restoration_sign_reclassification/verify_corrected_restoration_forest.py; mutation command: .venv/bin/python -B work/restoration_sign_reclassification/mutate_corrected_restoration_forest.py.
one-/two-port probe closure	schema/version: k2p-corrected-coherent-probe-closure-v1; authority: work/probe_coherence_corrected/probe_coherence_certificate.json; SHA-256: 93de7b0dd3aa581bdf12288eae8cb9ac42f20a9d9bb3eab35eee8ef9a759d390; producer command: .venv/bin/python -B work/probe_coherence_corrected/build_probe_coherence_corrected.py; replay command: .venv/bin/python -B work/probe_coherence_corrected/verify_probe_coherence_corrected.py; mutation command: .venv/bin/python -B work/probe_coherence_corrected/run_probe_coherence_mutations.py.
weak-sharpness named columns	schema/version: k2p-weak-sharpness-column-crosswalk-v1; authority: proof_compression_submission/analysis/WEAK_SHARPNESS_COLUMN_CROSSWALK.json; SHA-256: 827fc4a81b27da05916040e76e4de5b512c7a71e38577e495860a2cf63ee1af5; producer command: .venv/bin/python -B proof_compression_submission/analysis/build_weak_sharpness_column_crosswalk.py; replay command: .venv/bin/python -B proof_compression_submission/analysis/verify_weak_sharpness_column_crosswalk.py; mutation command: .venv/bin/python -B proof_compression_submission/analysis/test_weak_sharpness_column_crosswalk_mutations.py.
full-map certificate reseal	schema/version: k2p-full-map-domain-reseal-audit-v1; authority: work/final_theorem_release/full_map_reseal_audit.json; SHA-256: 8ec6b987422b0dd27a12d590e8aad4f2c34817e341024f0881000013baffc3df; producer/replay/mutation command: .venv/bin/python -B work/final_theorem_release/verify_full_map_reseal.py.
composite reseal differential	schema/version: k2p-composite-domain-reseal-diff-audit-v1; authority: work/final_theorem_release/composite_reseal_diff_audit.json; SHA-256: bc91fee3b7541fcae72c4db2e66776fbfc69c43890718239f0eea41bb2cc0654; producer/replay command: .venv/bin/python -B work/final_theorem_release/verify_composite_reseal_diff.py; mutation authority: the six fail-closed seal/domain mutations in the preceding reseal audit.

11 Frozen hash anchors

The following SHA-256 values are the compact anchors most useful for checking that the intended finite universe has been unpacked. The outer release lock authenticates all nested files and should remain the authority if any table is updated.

artifact	file SHA-256
unified release lock	58e32bd29f7a039e3da4e47398e32ee8277ad46cf62271a7ed80bf41688b18fb
raw-four composite ledger	431dac8898ad2a724d12c200687de1b377723e302214a79a11a03524a4084b96
raw-four independent replay	d44705d2266aa1360accec19cdb8bf0fea648fc9c029ae96adb437a04ad5cce10
raw-four terminal registry	0a1818655429d60660c1ed87f3fbe412701f386b081562b3a4caa54079069f1d
raw-four mutation report	449228fa4013234280434495e5c79468750de2785d5a2a2a8ad7d91bbdcae3a7
five-port θ_2 composite ledger	805fc7f5a3de9dad2c63a210208075cf19910cf811ffd08878f32782ce71b659
five-port independent replay	ed57b46c5e05a73f08a26afc99ae1bd5f0076e941b2da7ac411ecc52f3b63e58
five-port mutation report	c1c8012941ac58ae3f01a890caaa9e9ad05e8342d11a41d8a911de18dc3e6775
full-map certificate reseal audit	8ec6b987422b0dd27a12d590e8aad4f2c34817e341024f0881000013baffc3df
composite reseal differential audit	bc91fee3b7541fcae72c4db2e66776fbfc69c43890718239f0eea41bb2cc0654
corrected 997-parent forest	43bd2be5e7626a954fc4fa4cf45e8d0e6483c947ddc9cba80f2b1a13351bc3a8
restoration independent replay	24fa2e61f60610a8b24c4107ec7f866278f0cc671ca203d7aaa40a37bea291dd
probe input contract	7f686ae99dd5e6dafc1c04396b711d294a0bddd6a25574f9ea809b831ad7b377
full probe primary	93de7b0dd3aa581bdf12288eae8cb9ac42f20a9d9bb3eab35eee8ef9a759d390

artifact	file SHA-256
full probe independent graph audit	47f527dd56ce355d911c44ebc55e24a1e9c7f14f379810902b49ae068261e74f
full probe adversarial mutation report	b0df0584163150c9a823b4e364b8ee46c196ae8abb28fdca4d3d5893a97bfea7
weak-sharpness primary	e66c78a0aeab990b4dc448f4f064b37e1e15ecbff75a5f472bf116d4464378bd
weak-sharpness independent audit	cfd8d3a2ebc7431d141cac6ebe943e25730eb086fbc84b52833a40bee40a5d52

The combined one-/two-port ordered-ledger root is

7868fed6f8e0c10fcb9740da8ffdc7f64ea68939c99cba6f364da4cfd90bf50.

12 Replay protocol

Run from the project root. The locked environment uses Python 3.10 or newer; the qualified release used Python 3.14.6, NetworkX 3.5, and SymPy 1.14.0.

```
python3 -m venv .venv
.venv/bin/python -m pip install --upgrade pip
.venv/bin/python -m pip install \
  -r work/final_theorem_release/requirements.txt

.venv/bin/python -B \
  work/final_theorem_release/build_release_lock.py --check --require-ready
.venv/bin/python -B \
  work/final_theorem_release/verify_final_theorem_release.py --quick
.venv/bin/python -B \
  work/final_theorem_release/run_release_mutations.py
```

For full primitive regeneration:

```
.venv/bin/python -B \
  work/final_theorem_release/verify_final_theorem_release.py --full
```

The entry points reject optimized Python (`python -O`). Full mode regenerates large ledgers by streaming; it does not require an uncompressed multi-gigabyte temporary JSONL file. A detached clean checkout at commit `1e9ff6c6` completed the full replay on 22 August 2026 with all 35 layers passing, zero blockers, and `promotion_ready=true`. The measured wall time was 5,172.89 seconds (86.2 minutes); the verifier’s internal elapsed time was 5,172.248447 seconds. macOS `/usr/bin/time -l` reported maximum resident set size 1,960,001,536 bytes and peak memory footprint 491,504,408 bytes; these are distinct operating-system metrics. The byte-exact replay report and telemetry are `proof_compression_submission/output/FINAL_CLEAN_FULL_REPLAY.json` (SHA-256 `7939b389880de80b7d8abd69022e0b69d2dc4188815854b294d3384fa24c9e18`) and `proof_compression_submission/output/FINAL_CLEAN_FULL_REPLAY_TELEMETRY.json` (SHA-256 `8779854633d9a52ba3d7bc9278ccbcc3918e51987bb4c30204c0adcd9771ce16`).

13 Mutation coverage

The independent suites reject at least the following semantic faults:

- omitted or duplicated raw relation;
- swapped or false topology reason;
- false target rank upper bound or rank exclusion;
- reassignment of a terminal class;
- omitted restoration role or child;
- wrong source insertion, target attachment, or parent;
- restoration cycle or missing continuation layer;
- broken boundary transport or parent restriction;
- omitted probe anchor, pendant site, reticulation-incoming site, or root-suppressed site;
- splitting the two artificial root halves into distinct physical sites;
- source–target reversal omission;
- reassignment of quadratic, cubic, quartic, quintic, or $\mathcal{T}_i$ certificate;
- reintroduction of a rooted `triple_type` topology oracle;
- changed sharpness arrows, parents, inheritance complements, Fourier coordinates, rank minors, or incomplete rooting list; and
- optimized-Python execution that would disable fail-closed checks.

Mutation rejection demonstrates that the verifier watches these data. It is not a substitute for the direct mathematical implication associated with each certificate type.

14 Recent-source verification note

Citation metadata were checked on 21 August 2026 against primary publisher or authoritative preprint records where available:

source	verified locator and status
Kimura (1980) [?]	Springer version of record, https://doi.org/10.1007/BF01731581 , vol. 16, pp. 111–120.
Evans–Speed (1993) [?]	Project Euclid/Annals record, https://doi.org/10.1214/aos/1176349030 , vol. 21, pp. 355–377.
Sturmfels–Sullivant (2005) [?]	author/arXiv and journal metadata agree on J. Comput. Biol. 12(2), pp. 204–228, https://doi.org/10.1089/cmb.2005.12.204 .
Gross et al. (2021) [?]	Springer version of record, https://doi.org/10.1007/s00285-021-01653-8 , J. Math. Biol. 83:32.
Cox–Gross–Martin (2025) [?]	Springer version of record, https://doi.org/10.1007/s11538-025-01506-1 , Bull. Math. Biol. 87:132.
Ardiyansyah (2021) [?]	arXiv primary record https://arxiv.org/abs/2104.12479v1 ; the abstract states the simple/semisimple level-2 invariant scope used in the related-work summary.
Holtgreffe et al. (2025) [?]	Springer version of record, https://doi.org/10.1007/s11538-025-01549-4 , Bull. Math. Biol. 87:168; the publisher records a 14 January 2026 correction.
Holtgreffe et al. (2026) [?]	Springer version of record, https://doi.org/10.1007/s12064-025-00453-8 , Theory Biosci. 145:4.
Huber et al. (2025) [?]	Springer version of record, https://doi.org/10.1007/s11538-025-01510-5 , Bull. Math. Biol. 87:136; Lemma 4.2 is the two-generator result used here.
Englander et al. (2025) [?]	bioRxiv DOI https://doi.org/10.1101/2025.04.18.649493 ; theorem numbering and the 4 July 2026 version-4 content were checked against the locally archived source XML/PDF.
Brits et al. (2026) [?]	arXiv primary record https://arxiv.org/abs/2607.12919v2 ; version 2 dated 29 July 2026.
Kriebel (2026) [?]	versioned public companion JC repository release https://github.com/AlecKriebel/Math/releases/tag/stc-jc-sharp-boundary-v1.1.4 , source commit b5b855f5e6552e1d87f32d90851ea2def5330364 ; no DOI is claimed.

The local K2P/K3P theta-collision manuscript is provisional and has no verified public bibliographic status in this draft. It is therefore not cited as a published result or used as theorem evidence.

15 Pending submission metadata and licenses

The following items require explicit human decisions before submission:

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The mutable public repository is <https://github.com/AlecKriebel/Math>; the project lives under `k2p_level2_identifiability_closure`. No email address, funding statement, competing-interests

declaration, or license has been inferred from older project files. This draft asserts none of them. A DOI-bearing release should be created only for a genuinely final immutable submission package.

Disclosure

Generative AI assisted with discovery, code, drafting, and adversarial review. The cited independent replays are separately implemented programs, not independent human review. The author accepts responsibility for the final mathematics, software, and submission.

References